

PHASE 5 – PROJECT DEVELOPMENT

Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders

1. Introduction

The Project Development Phase is the implementation stage of the project. In this phase, the system design and requirements are converted into a working application.

2. Development Environment

The following tools and technologies can be used for development:

- Code Editor: Visual Studio Code
- Programming Language: Based on the selected project technology
- Database: Selected database system
- Version Control: Git
- Repository: GitHub
- Browser: Google Chrome / Microsoft Edge

3. System Development

The application is developed by implementing the main modules identified during the design phase.

User Management

- User registration and login.
- Role-based access.
- User profile management.

Laptop Catalog

- Display available standard laptop models.
- Show laptop specifications.
- Allow users to select a suitable laptop.

Laptop Request

- Provide a laptop request form.
- Collect employee and laptop details.
- Submit the request for approval.

Approval Management

- Display pending requests to authorized managers.
- Allow approval or rejection.
- Update request status automatically.

Procurement Management

- Display approved requests.
- Verify laptop requirements.
- Process the order.
- Update procurement status.

Request Tracking

Users can track the progress of their laptop request through different statuses.

Submitted → Pending Approval → Approved → IT Review → In Procurement → Ordered → Completed

4. Database Development

The database is created to store and manage important project information such as:

- User details.
- Laptop details.
- Laptop requests.
- Approval information.
- Procurement and order details.

Proper relationships are maintained between the database entities.

5. Validation and Security

The system includes basic validation to ensure that required information is entered correctly.

Examples:

- Required fields cannot be left empty.
- Only authorized users can approve requests.

- User information is protected through authentication.
- Invalid data is rejected before submission.

6. Integration

All major modules are integrated to create a complete workflow.

User Login → Laptop Request → Approval → IT Review → Procurement → Order Completion

The database is connected with the application to store and retrieve project information.

7. Version Control

Git and GitHub are used to:

- Store project source code.
- Track changes.
- Maintain project versions.
- Upload phase-wise project documents.
- Provide the required public repository.

8. Development Output

At the end of this phase, the following are expected:

- Working user interface.
- Laptop request module.

- Approval workflow.
- Procurement module.
- Database integration.
- Request tracking.
- Complete application workflow.

9. Conclusion

The Project Development Phase converts the planned design into a working IT procurement application. The developed system provides a structured process for submitting, approving, processing, and tracking standard laptop orders.